



Electronic I

Lecture 10

Methods of Analysis:

Nodal analysis

Assistant Lecturer. Saif Almansouri

Al-Nahrain University / College of Engineering

Department of Computer Engineering

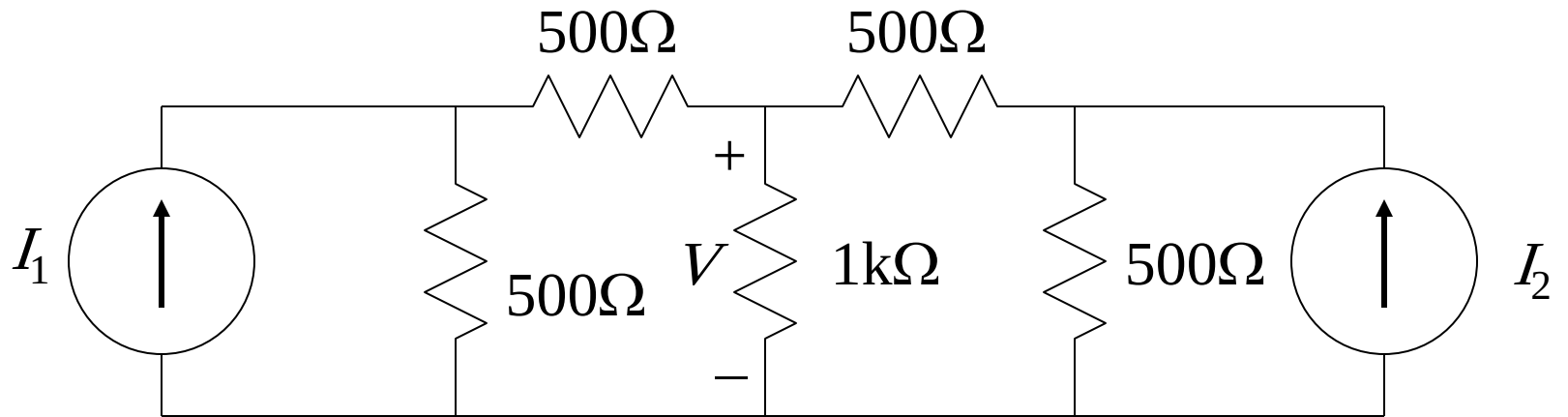
First Year / Spring Semester

safe.abduljassim@eng.nahrainuniv.edu.iq

Example: A Summing Circuit

- The output voltage V of this circuit is proportional to the **sum** of the two input currents I_1 and I_2 .
- This circuit could be useful in audio applications or in instrumentation.
- The output of this circuit would probably be connected to an amplifier.

Summing Circuit



This One!

- There are no series or parallel resistors to combine.
- We do not have a single loop or a double node circuit.
- We need a more powerful analysis technique:

Nodal Analysis

Why Nodal Analysis?

- The analysis techniques (voltage divider, equivalent resistance, etc.) provide an intuitive approach to analyzing circuits.
- They cannot analyze all circuits.
- They cannot be easily automated by a computer.

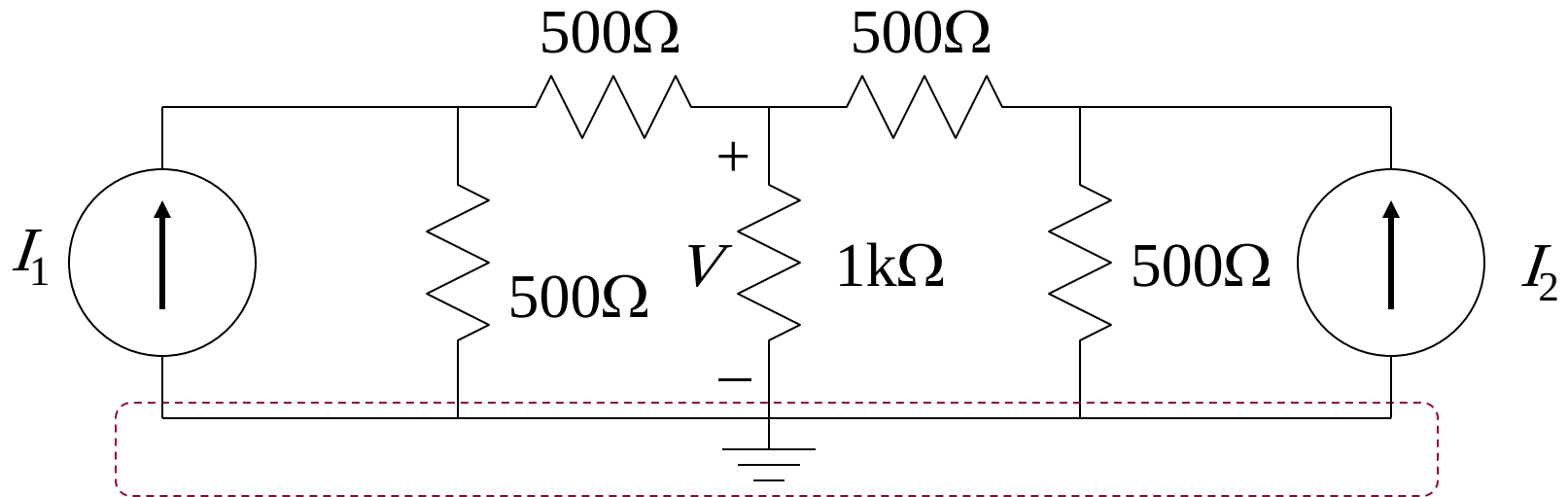
Node and Loop Analysis

- Node analysis and loop analysis are both circuit analysis methods which are systematic and apply to most circuits.
- Analysis of circuits using node or loop analysis requires solutions of systems of linear equations.
- These equations can usually be written by inspection of the circuit.

Steps of Nodal Analysis

- 1. Choose a reference node.**
2. Assign node voltages to the other nodes.
3. Apply KCL to each node other than the reference node; express currents in terms of node voltages.
4. Solve the resulting system of linear equations.

Reference Node



The reference node is called the *ground* node.

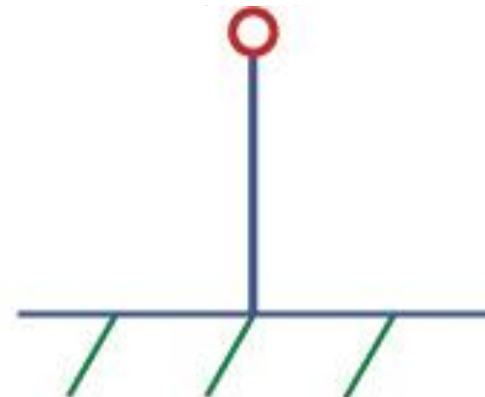
Common symbols for indicating a reference node,
(a) common ground, (b) ground, (c) chassis.



(a)



(b)

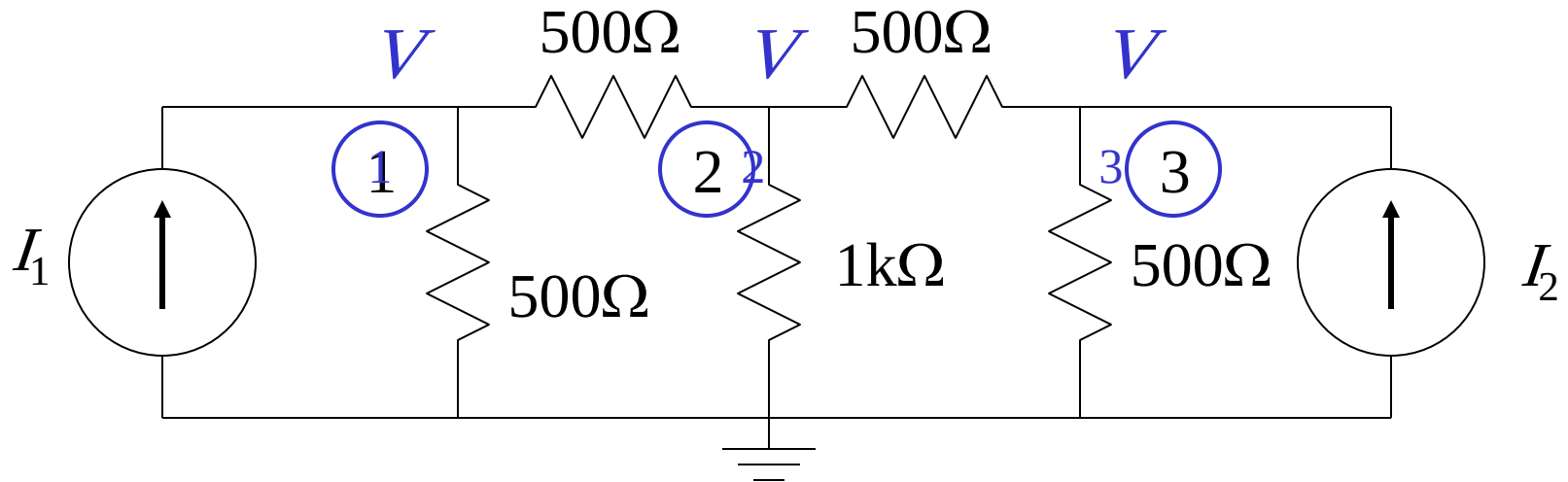


(c)

Steps of Nodal Analysis

1. Choose a reference node.
- 2. Assign node voltages to the other nodes.**
3. Apply KCL to each node other than the reference node; express currents in terms of node voltages.
4. Solve the resulting system of linear equations.

Node Voltages

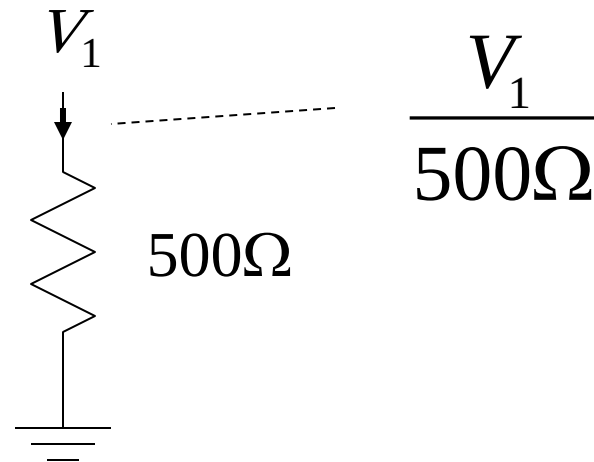
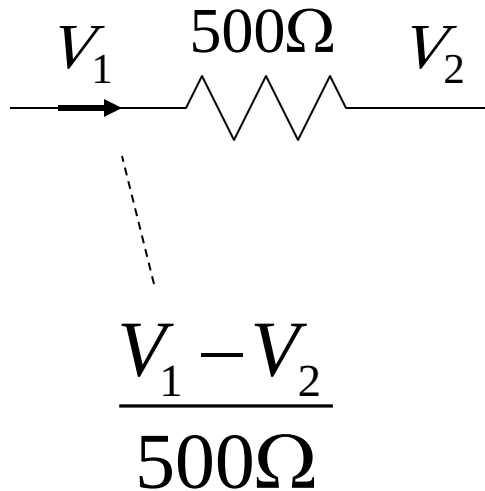


V_1 , V_2 , and V_3 are unknowns for which we solve using KCL.

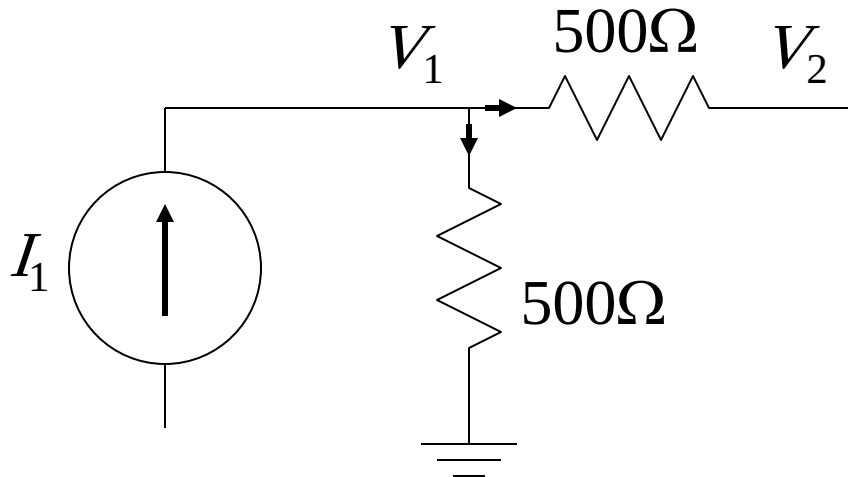
Steps of Nodal Analysis

1. Choose a reference node.
2. Assign node voltages to the other nodes.
- 3. Apply KCL to each node other than the reference node; express currents in terms of node voltages.**
4. Solve the resulting system of linear equations.

Currents and Node Voltages

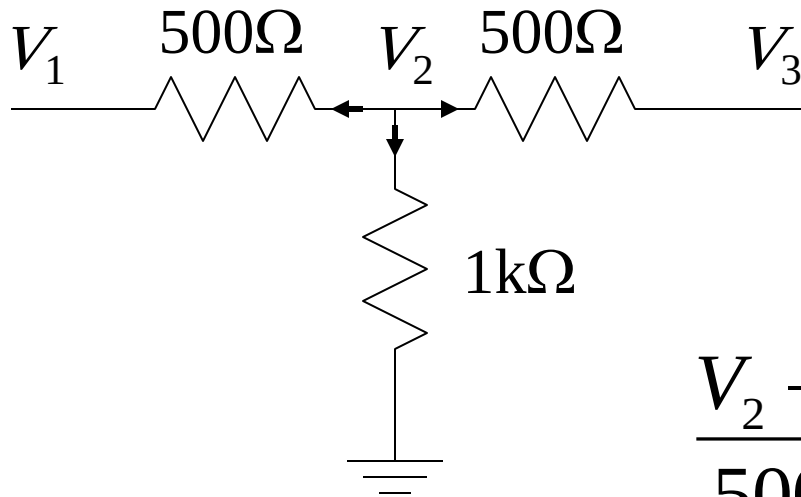


KCL at Node 1



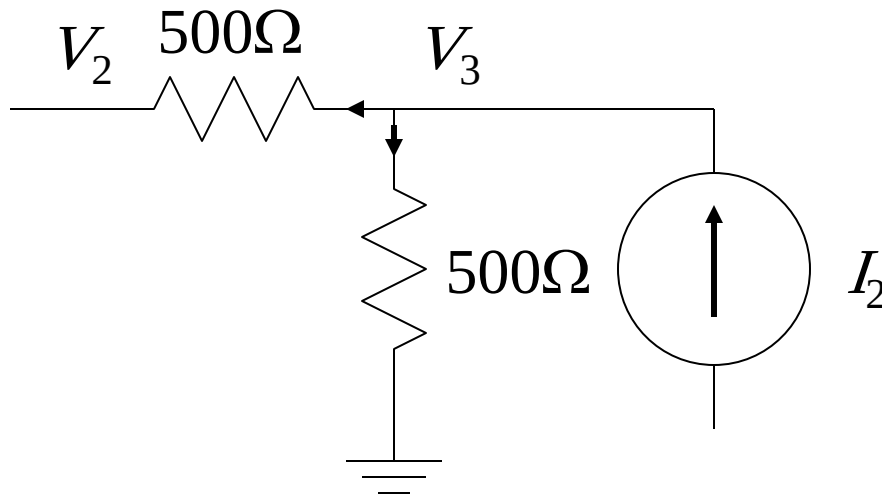
$$I_1 = \frac{V_1 - V_2}{500\Omega} + \frac{V_1}{500\Omega}$$

KCL at Node 2



$$\frac{V_2 - V_1}{500\Omega} + \frac{V_2}{1\text{k}\Omega} + \frac{V_2 - V_3}{500\Omega} = 0$$

KCL at Node 3



$$\frac{V_3 - V_2}{500\Omega} + \frac{V_3}{500\Omega} = I_2$$

Steps of Nodal Analysis

1. Choose a reference node.
2. Assign node voltages to the other nodes.
3. Apply KCL to each node other than the reference node; express currents in terms of node voltages.
- 4. Solve the resulting system of linear equations.**

System of Equations

- Node 1:

$$V_1 \left(\frac{1}{500\Omega} + \frac{1}{500\Omega} \right) - \frac{V_2}{500\Omega} = I_1$$

- Node 2:

$$-\frac{V_1}{500\Omega} + V_2 \left(\frac{1}{500\Omega} + \frac{1}{1k\Omega} + \frac{1}{500\Omega} \right) - \frac{V_3}{500\Omega} = 0$$

System of Equations

- Node 3:

$$-\frac{V_2}{500\Omega} + V_3\left(\frac{1}{500\Omega} + \frac{1}{500\Omega}\right) = I_2$$

Equations

- These equations can be written by inspection.
- The left side of the equation:
 - The node voltage is multiplied by the sum of *conductances* of all resistors connected to the node.
 - Other node voltages are multiplied by the conductance of the resistor(s) connecting to the node and subtracted.

Equations

- The right side of the equation:
 - The right side of the equation is the sum of currents from sources entering the node.

Matrix Notation

- The three equations can be combined into a single matrix/vector equation.

$$\begin{bmatrix} \frac{1}{500\Omega} + \frac{1}{500\Omega} & -\frac{1}{500\Omega} & 0 \\ -\frac{1}{500\Omega} & \frac{1}{500\Omega} + \frac{1}{1\text{k}\Omega} + \frac{1}{500\Omega} & -\frac{1}{500\Omega} \\ 0 & -\frac{1}{500\Omega} & \frac{1}{500\Omega} + \frac{1}{500\Omega} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} I_1 \\ 0 \\ I_2 \end{bmatrix}$$

Matrix Notation

- The equation can be written in matrix-vector form as

$$\mathbf{A}\mathbf{v} = \mathbf{i}$$

- The solution to the equation can be written as

$$\mathbf{v} = \mathbf{A}^{-1} \mathbf{i}$$

Advantages of Nodal Analysis

- Solves directly for node voltages.
- Current sources are easy.
- Voltage sources are either very easy or somewhat difficult.
- Works best for circuits with few nodes.
- Works for any circuit.

Advantages of Loop Analysis

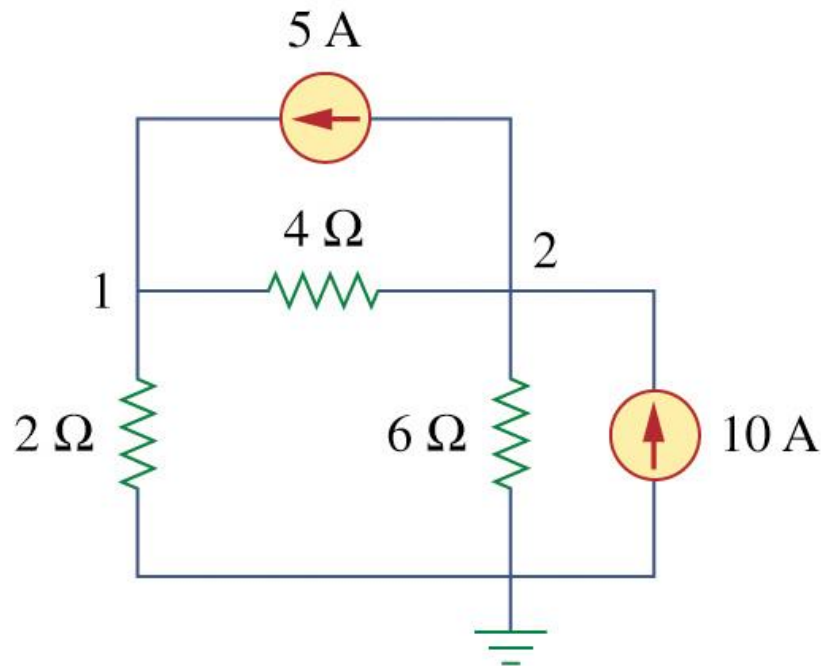
- Solves directly for some currents.
- Voltage sources are easy.
- Current sources are either very easy or somewhat difficult.
- Works best for circuits with few loops.

Disadvantages of Loop Analysis

- Some currents must be computed from loop currents.
- Does not work with non-planar circuits.
- Choosing the supermesh may be difficult.

Example

- Calculate the node voltage in the circuit shown in Fig.



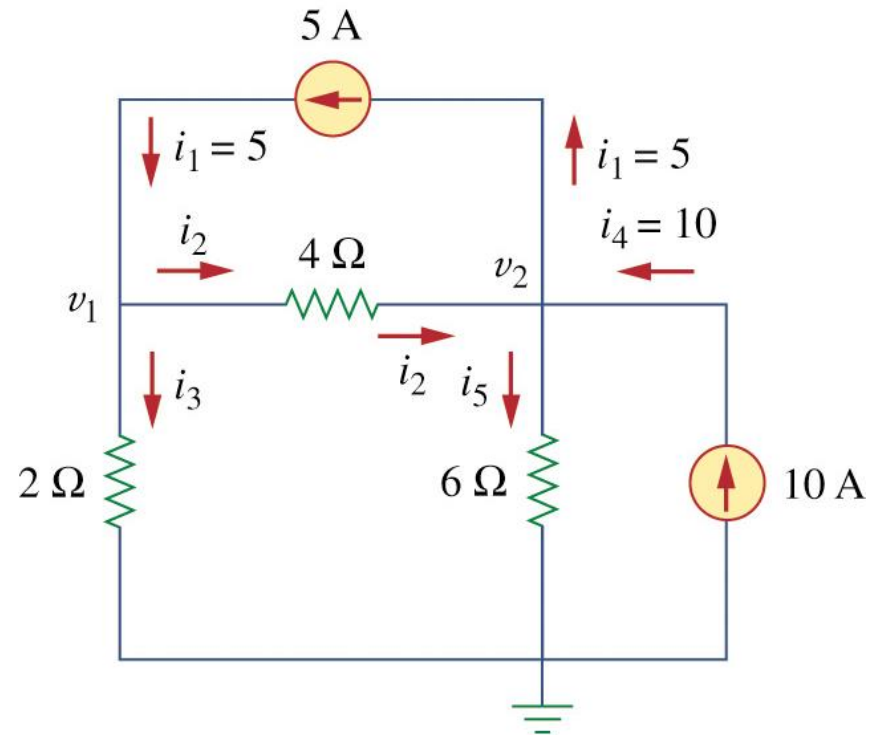
(a)

Example

- At node 1

$$i_1 = i_2 + i_3$$

$$\Rightarrow 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$



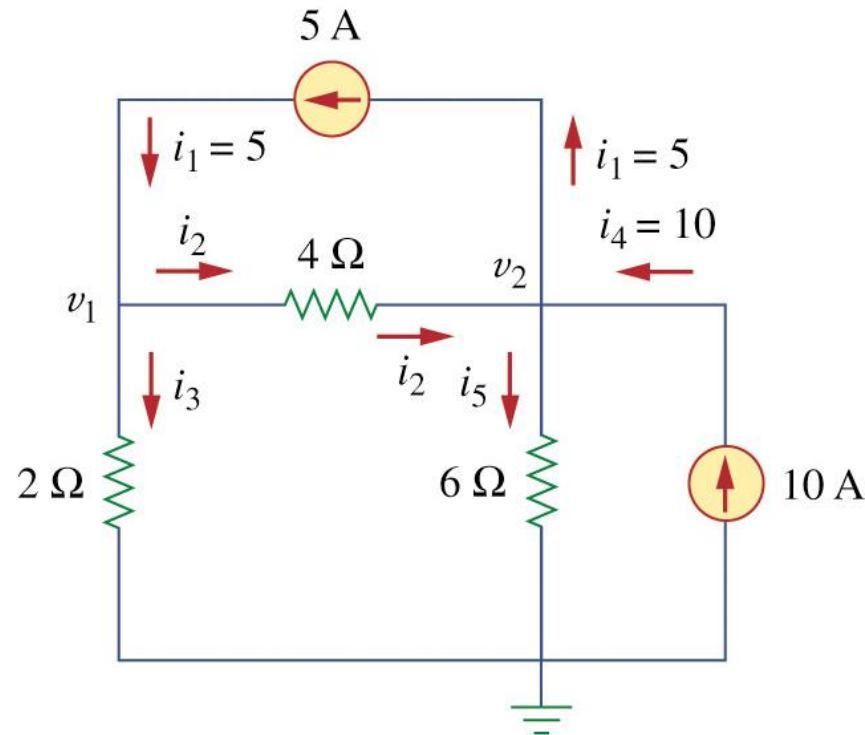
(b)

Example

- At node 2

$$i_2 + i_4 = i_1 + i_5$$

$$\Rightarrow 5 = \frac{v_2 - v_1}{4} + \frac{v_2 - 0}{6}$$



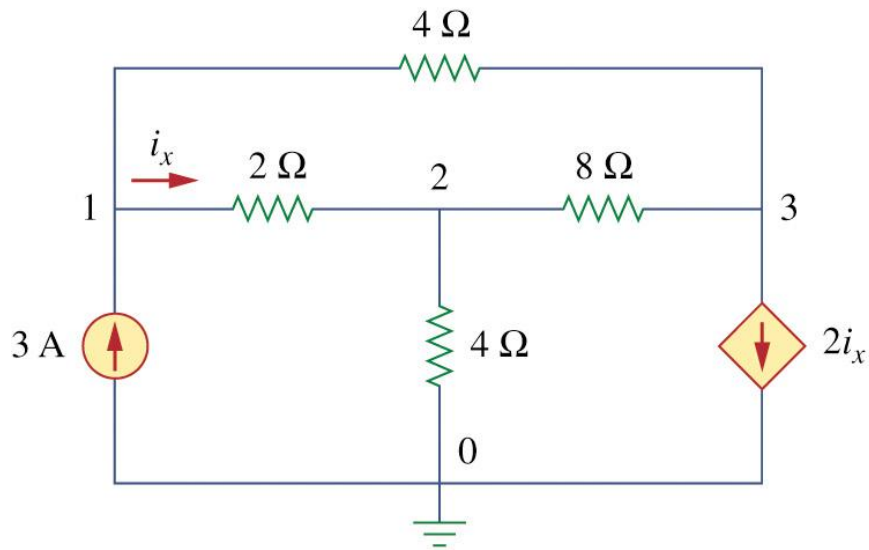
(b)

- In matrix form:

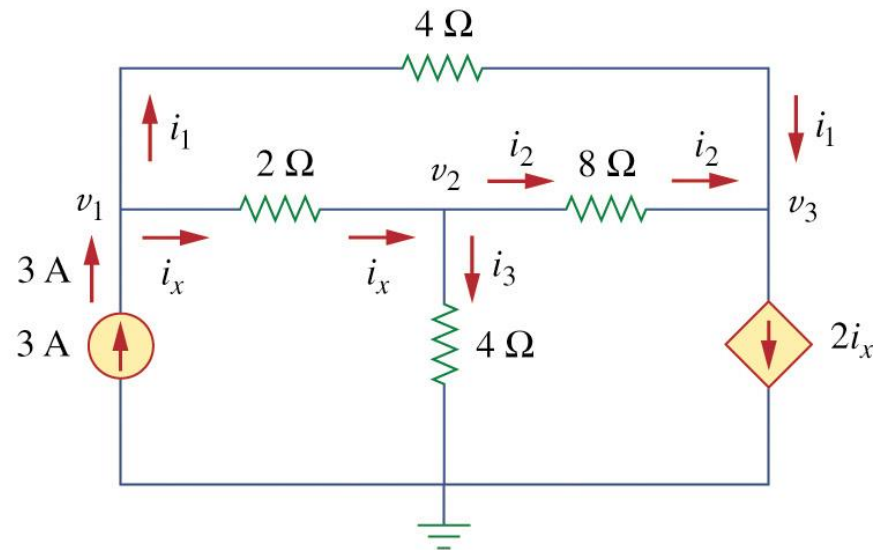
$$\begin{bmatrix} \frac{1}{2} + \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{6} + \frac{1}{4} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

Example

- Determine the voltage at the nodes in Fig.



(a)

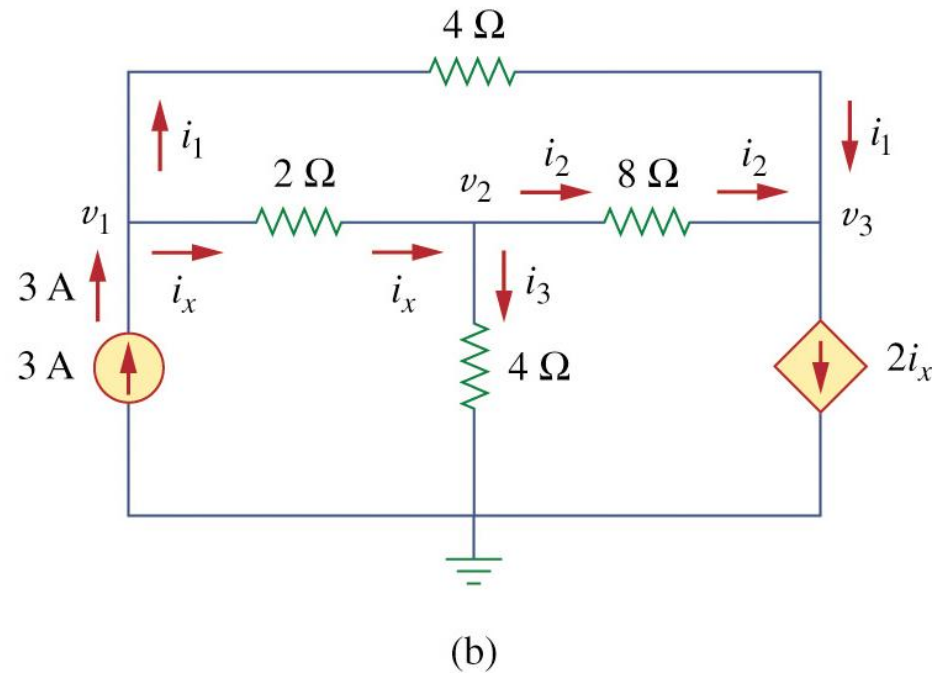


(b)

- At node 1,

$$3 = i_1 + i_x$$

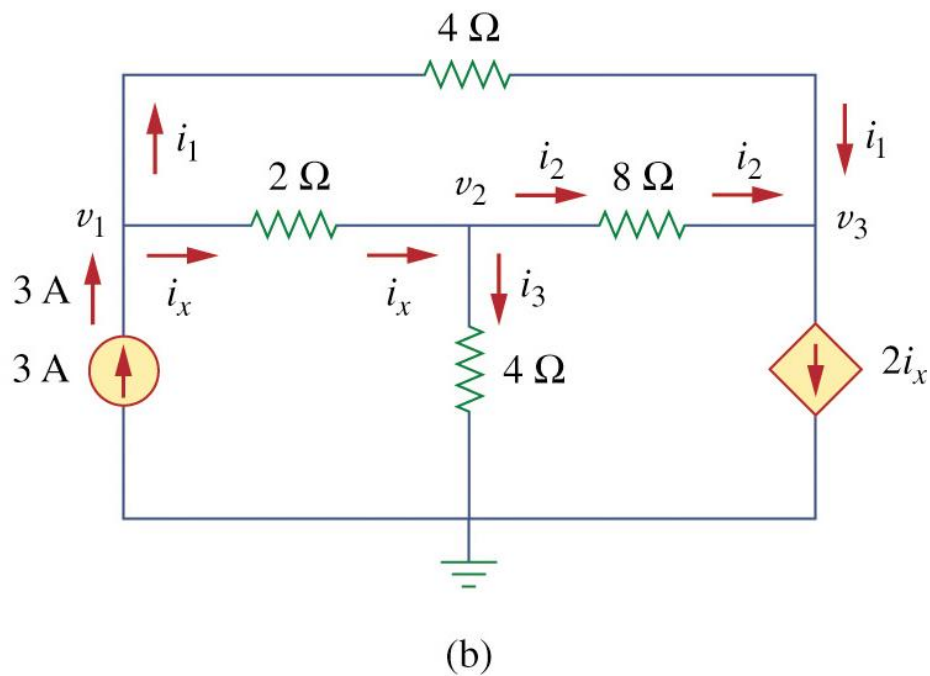
$$\Rightarrow 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$$



- At node 2

$$i_x = i_2 + i_3$$

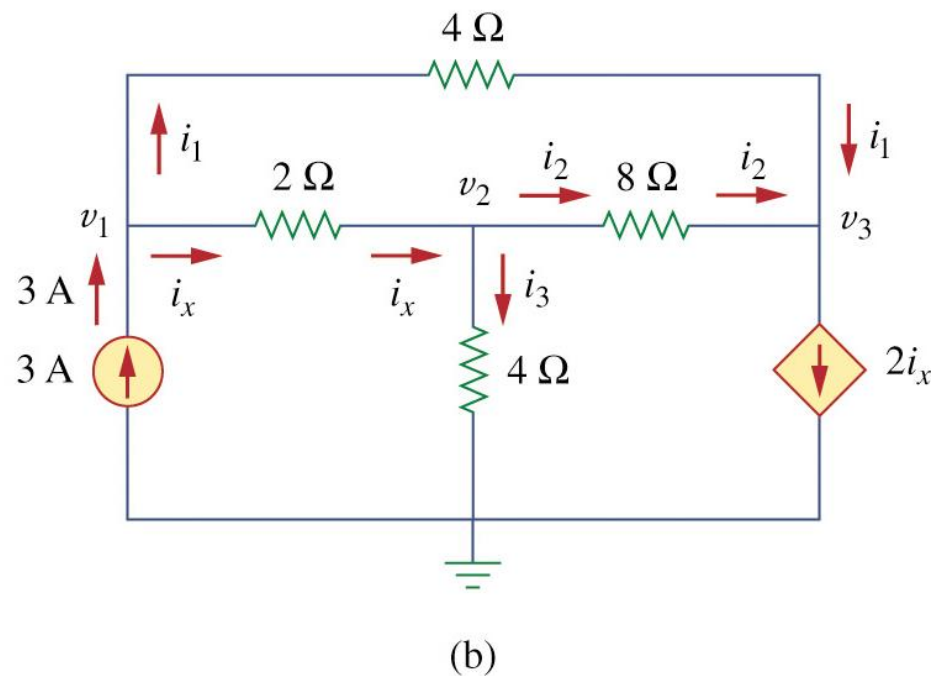
$$\Rightarrow \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$$



- At node 3

$$i_1 + i_2 = 2i_x$$

$$\Rightarrow \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$$



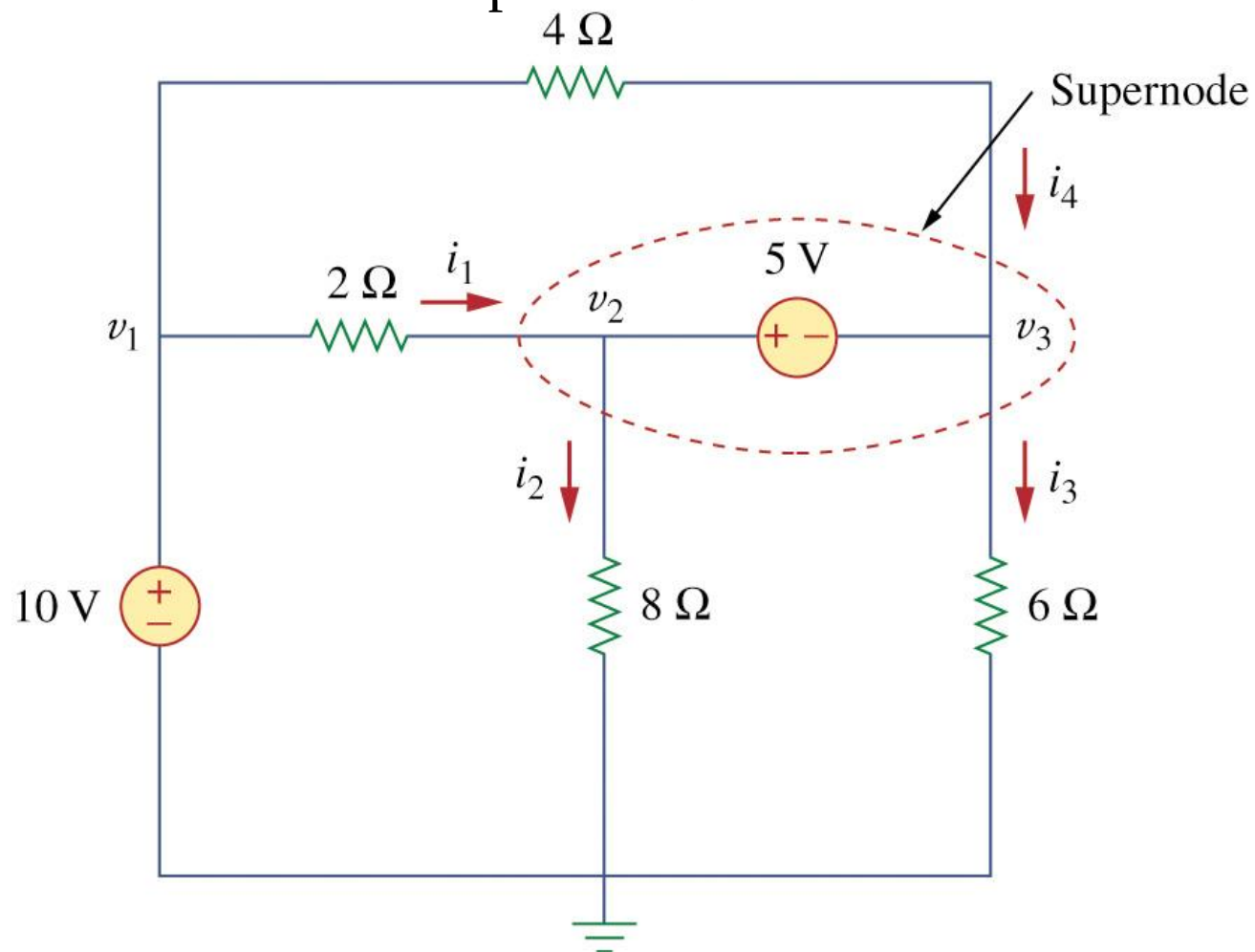
Example

- In matrix form:

$$\begin{bmatrix} \frac{3}{4} & -\frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & \frac{7}{8} & -\frac{1}{8} \\ \frac{3}{4} & -\frac{9}{8} & \frac{3}{8} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

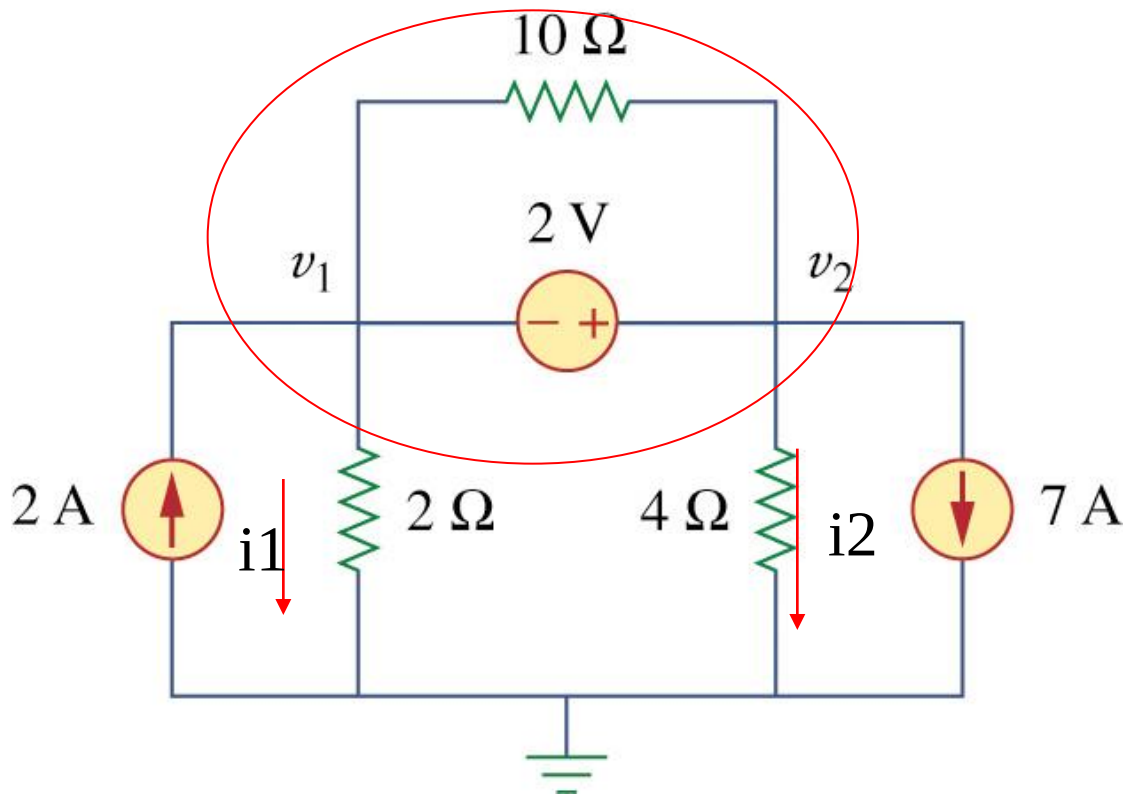
- A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.
- The required two equations for regulating the two nonreference node voltages are obtained by the KCL of the supernode and the relationship of node voltages due to the voltage source.

A circuit with a supernode.



Example

- For the circuit shown in Fig. find the node voltages.



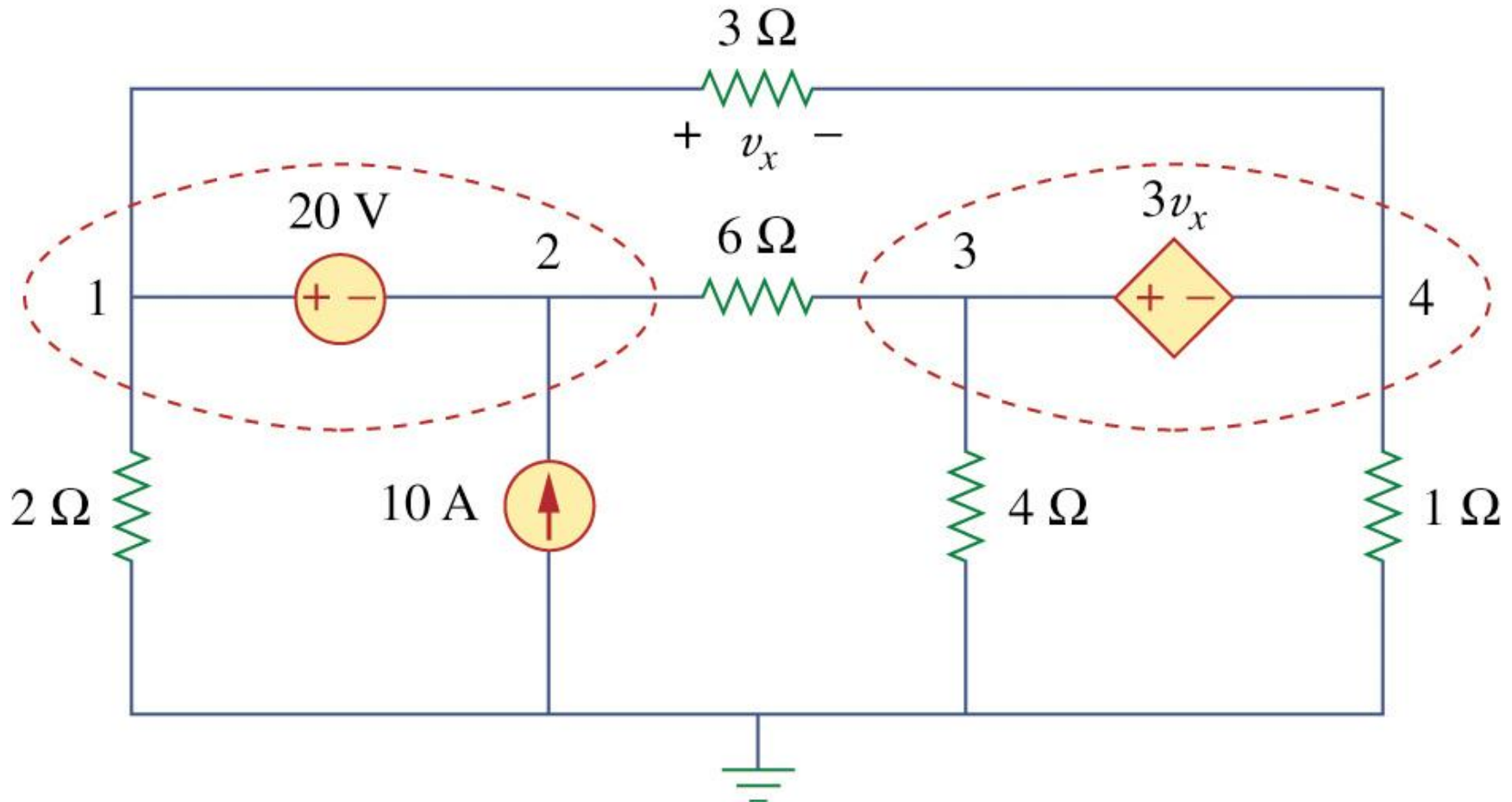
$$2 - 7 - i_1 - i_2 = 0$$

$$2 - 7 - \frac{v_1}{2} - \frac{v_2}{4} = 0$$

$$v_1 - v_2 = -2$$

Example

Find the node voltages in the circuit of Fig. 3.12.

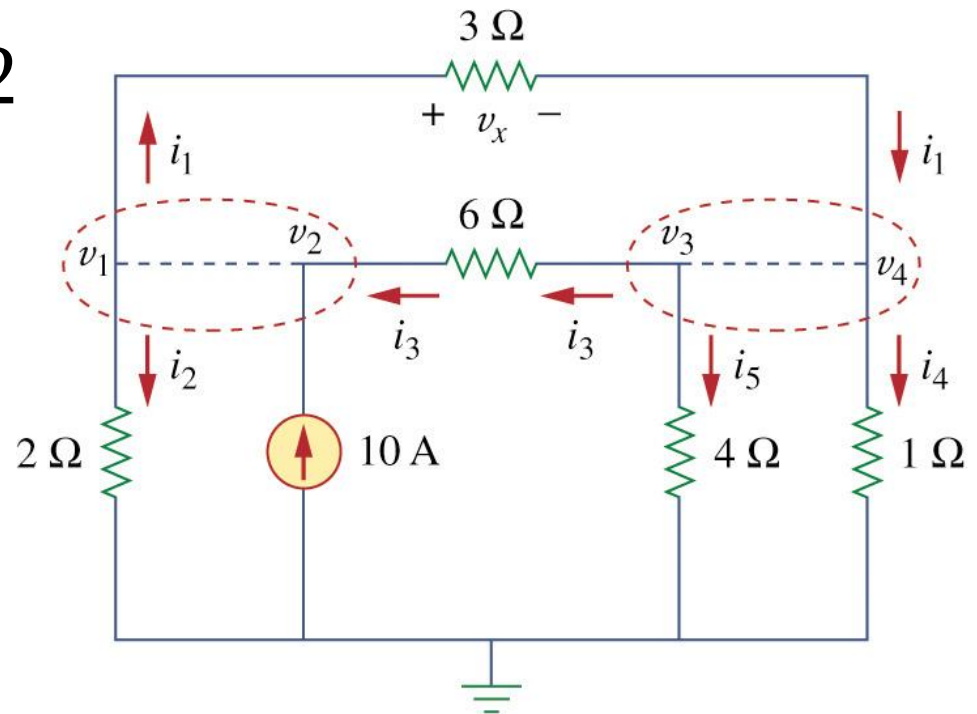


Example

- At supernode 1-2

$$\frac{v_3 - v_2}{6} + 10 = \frac{v_1 - v_4}{3} + \frac{v_1}{2}$$

$$v_1 - v_2 = 20$$



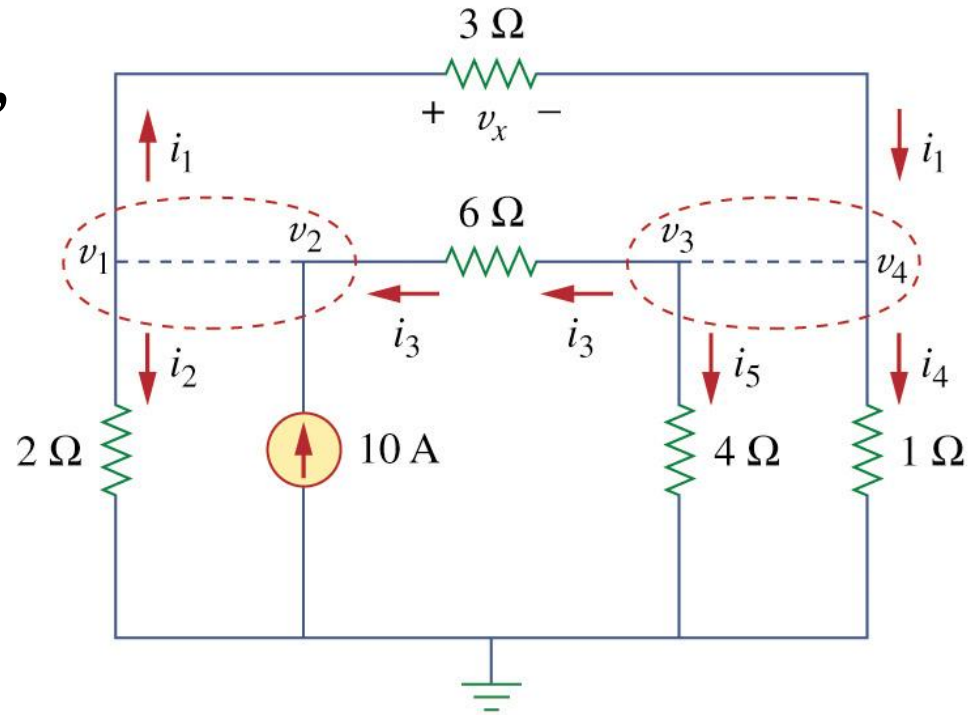
(a)

Example

- At supernode 3-4,

$$\frac{v_1 - v_4}{3} = \frac{v_3 - v_2}{6} + \frac{v_4}{1} + \frac{v_3}{4}$$

$$v_3 - v_4 = 3(v_1 - v_4)$$



(a)

